



Protocol name: RP02 Preparation of Reynold's Lead Citrate ^[1]

Version: 1.1

Status: Approved

Date Approved: 03/04/2024

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Keywords: Post-staining; Positive staining

Health & Safety requirements

Ensure you are familiar with all relevant local Health and Safety documents and procedures (Material Safety Data Sheets, Procedural Risk Assessments, Standard Operating Procedures etc.) before starting.

Safety critical: All procedures designated safety critical must be performed in a fume hood (where appropriate) and the following PPE must be worn - lab coat, gloves, and eye protection.

Procedural Risk Assessments: Handling of lead nitrate, sodium hydroxide, lead citrate

Materials

- ☐ Balance
- ☐ Magnetic stirrer plate
- ☐ Magnetic stirring bars
- ☐ 3 ml plastic Pasteur pipettes SLS PIP4206
- ☐ Long neck volumetric flask, 50ml
- ☐ Measuring cylinders
- ☐ pH meter
- ☐ Fridge, 4°C

Chemicals

- ☐ ddH₂O; double distilled water
- ☐ Lead nitrate [Pb(NO₃)₂] Sigma-Aldrich 228621
- ☐ Sodium citrate [Na₃C₆H₅O₇·2H₂O] BDH 10242
- ☐ Sodium hydroxide [NaOH] 1 N Fisher Scientific S/4920/53

Procedure

Note: Lead citrate is used as a positive stain and the below protocol was developed by Edward Reynolds (1963) ^[1]. Lead citrate is unique as it uses citrate as a chelating agent. Citrate is added in excess of lead to prevent the lead combining with carbonate and forming contaminants. The excess citrate does

not appear to interfere with the binding of lead citrate with tissue. Anionic tissue binding sites appear to have a greater affinity for lead citrate than citrate.

Preparing Reynold's lead citrate

1. Boil 100 ml of double distilled water to remove dissolved carbon dioxide. Leave to cool.

Note: Use this pre-boiled ddH₂O for all subsequent steps.

2. Prepare 1 N solution of NaOH. **Safety critical**
 - a. Weigh 0.5 g of NaOH in a flask.
 - b. Add 12.5 ml of ddH₂O.
 - c. Add a clean magnetic stirrer and place on a stirrer until dissolved.

Sodium Hydroxide	State:	Hazard Codes:	
	Solid	H290, H314	

3. Weigh 1.33 g of lead nitrate in a weigh boat and add to the 50 ml long neck volumetric flask (as shown in figure 1). **Safety critical**

Lead Nitrate	State:	Hazard Codes:	
	Solid	H272, H302, H332, H318, H360Df, H373, H410	



Figure 1. A 50ml long neck volumetric flask.

4. Weigh 1.76 g of sodium citrate in a weigh boat and add this to your long neck volumetric flask.
5. Add 30 ml of ddH₂O. Place a cap on the flask and seal with parafilm. Shake the volumetric flask vigorously for 60 seconds at regular intervals over 30 minutes.

6. Add 5 ml of ddH₂O to the volumetric flask. The solution will be cloudy.
7. pH the solution to 12.0 ± 0.1
 - a. First check the pH of the solution.
 - b. Then slowly add the 1 N solution of NaOH in a dropwise manner until the desired pH is reached. Note that 2-3 ml of NaOH can be added before coming close to the final pH, when caution should be taken since the pH will change very quickly. Pay close attention to the opaqueness of the solution - it will become clear close to the final pH. If you overshoot the pH then the solution should be discarded.
8. Add ddH₂O to make up to a total volume of 50 ml. **Safety critical**

Lead Citrate	State: Liquid	Hazard Codes: H302, H332, H360Df, H373, H410	
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Note: Reynold's lead citrate should be stored in glass or polyethylene bottles. The staining solution is stable for a period of 6 months ^[2].

References:

- [1] Reynolds, E. (1963) 'The use of lead citrate at high pH as an electron-opaque stain in electron microscopy.', *The Journal of cell biology*, 17(1), pp. 208-212.
- [2] Hyatt, M.A. (2000) *Principles and Techniques of Electron Microscopy Biological Applications (Fourth Edition)*. Cambridge University Press. 299-300.